

Act Structure

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1 Introduction

The beats of a structured timeline are a tree. The tree structure technically accommodates any depth of tree, but we opted to model only up to three layers and to call them "Acts", "Chapters" and "Scenes" by default.

e.g. suppose you opted for three layers; you might add two Acts. You can add Chapters to those Acts independently so that Act I has two Chapters and Act II has three Chapters. Continuing in this way, you can add Scenes to each of those Chapters and might end up with three scenes in Act II's second Chapter, for example.

2 Customising Structure

There are controls to customize the depth of the hierarchy and each levels:

- Default beat name (you can change "Act" to "Top" from the default three-level structure and then top level, unnamed chapters will be called "Top 1", "Top 2" etc.).
- Text colour – the colour of the beat titles.
- Text size – the size of the beat title text.
- Border colour – the colour of borders drawn around beat titles.
- Border style – "Dashed", "Solid", "Dotted" and "None" (to be drawn around beat titles).
- Background colour – the colour to draw behind beat titles.

NOTE: The structure configuration can differ per book in a project.

2.1 Selectors

`sortedHierarchyLevels` produces a list of hierarchy levels sorted by their level. The following source listing is an example of what it produces.

```
[
  {
    "name": "Act",
    "level": 0,
    "autoNumber": true,
    "textSize": 24,
    "borderStyle": "SOLID",
    "backgroundColor": "none",
    "textColor": "#e5554f",
    "borderColor": "#e5554f",
    "dark": {
      "textColor": "#ffb8b5",
      "borderColor": "#ffb8b5"
    },
    "light": {
      "textColor": "#e5554f",
      "borderColor": "#e5554f"
    }
  },
  {
    "name": "Chapter",
    "level": 1,
    "autoNumber": true,
    "textSize": 24,
    "borderStyle": "DASHED",
    "backgroundColor": "none",
    "textColor": "DarkGoldenrod",
    "borderColor": "#78be20",
    "dark": {
      "textColor": "#baed79",
      "borderColor": "#baed79"
    },
    "light": {
      "textColor": "DarkGoldenrod",
      "borderColor": "#78be20"
    }
  }
]
```

```

    }
  },
  {
    "name": "Scene",
    "level": 2,
    "autoNumber": true,
    "textSize": 24,
    "borderStyle": "NONE",
    "backgroundColor": "none",
    "textColor": "#0b1117",
    "borderColor": "#6cace4",
    "dark": {
      "borderColor": "#ccc",
      "textColor": "#ccc"
    },
    "light": {
      "borderColor": "#6cace4",
      "textColor": "#0b1117"
    }
  }
]

```

2.2 Actions

The actions and selectors available here are basic. We didn't have a lot of time to improve on their design. Please feel free to update them.

2.2.1 Setting All Hierarchy Levels

You can specify a new structure for the current timeline with the action: `setHierarchyLevels` in the `hierarchy actions` module. It accepts an array of all three levels and dispatches an event to update the configuration for the current timeline.

Example usage:

```

setHierarchyLevels([
  {
    "name": "Act",
    "level": 0,
    "autoNumber": true,
    "textSize": 24,
    "borderStyle": "SOLID",
    "backgroundColor": "none",
    "textColor": "#e5554f",
    "borderColor": "#e5554f",
    "dark": {
      "textColor": "#ffb8b5",
      "borderColor": "#ffb8b5"
    },
    "light": {
      "textColor": "#e5554f",
      "borderColor": "#e5554f"
    }
  },
  {
    "name": "Chapter",
    "level": 1,
    "autoNumber": true,

```

```

    "textSize": 24,
    "borderStyle": "DASHED",
    "backgroundColor": "none",
    "textColor": "DarkGoldenrod",
    "borderColor": "#78be20",
    "dark": {
      "textColor": "#baed79",
      "borderColor": "#baed79"
    },
    "light": {
      "textColor": "DarkGoldenrod",
      "borderColor": "#78be20"
    }
  },
  {
    "name": "Scene",
    "level": 2,
    "autoNumber": true,
    "textSize": 24,
    "borderStyle": "NONE",
    "backgroundColor": "none",
    "textColor": "#0b1117",
    "borderColor": "#6cace4",
    "dark": {
      "borderColor": "#ccc",
      "textColor": "#ccc"
    },
    "light": {
      "borderColor": "#6cace4",
      "textColor": "#0b1117"
    }
  }
]
])

```

NOTE: the reducer does some work behind the scenes to ensure that the hierarchy levels are indexed by the numbers zero through 2 if they had invalid levels in the supplied array.

2.2.2 Editing a Single Hierarchy Level

You can modify a specific hierarchy level with the `editHierarchyLevel` action. It takes a single hierarchy level and swaps it in place of the existing hierarchy level (it determines which level to edit using the value of `level`).

Example usage:

```

editHierarchyLevel([
  {
    "name": "Act",
    "level": 0,
    "autoNumber": true,
    "textSize": 24,
    "borderStyle": "SOLID",
    "backgroundColor": "none",
    "textColor": "#e5554f",
    "borderColor": "#e5554f",
    "dark": {
      "textColor": "#ffb8b5",
      "borderColor": "#ffb8b5"
    },
    "light": {

```

```

        "textColor": "#e5554f",
        "borderColor": "#e5554f"
    }
}
1)

```

3 Rendering Structured Timelines

Plottr Desktop and Web supply users with three ways to visualise and manipulate structured timelines:

- **Default:** a view where Acts and Chapters can be individually collapsed into single columns and we draw stacks of cards where cards would be hidden by collapsing a column.
- **Tabbed:** a view where you don't get to see cards associated with the top level of hierarchy (Acts when there are three levels). Those beats are instead rendered as tabs and their Chapters and Scenes are rendered using the "Default" style inside of each tab.
- **Stacked:** a view where you don't get to see cards associated with Acts or Chapters. You only see the lowest level's cards. Chapters and Acts are rendered as bars above the Scenes that they contain.

3.1 Relevant Selectors

The following selectors produce the information needed to draw each of these views.

- **timelineViewSelector:** produces one of the strings: "default", "tabbed" or "stacked".
- **timelineViewIsStackedSelector:** produces true if the timeline view is set to "stacked".
- **timelineViewIsTabbedSelector:** produces true if the timeline view is set to "tabbed".
- **timelineActiveTabSelector:** produces the beat id that corresponds to the tab that's currently open in the tabbed view. e.g. the second tab might have been created from the beat with id 35. If the second tab was selected, this selector will produce 35.
- **timelineSparseBeatMap:** produces an object whose keys are the zero-indexed column indices of the timeline's columns and whose values are the beats that belong in those columns. It accounts for the current view and whether any beats are collapsed.
 e.g. if we're in the stacked view and there are three tiers, the beat ids for Acts and Chapters won't appear in the object's values.
 e.g. if we're in the tabbed view, the beat ids corresponding to tabs won't appear in the object's values.

Listing Sparse Beat Map is an example value that the selector might produce (the timeline view was tabbed). This object indicates that column 0 in the timeline table ought to contain beat 34's cards, column 1 ought to contain beat 20's cards etc.

- **searchedCardMetaDataMapSelector**: produces an object whose keys are the stringified coordinates (in the form ‘`#{line-id}-#{beat-id}`’ of all positions on the timeline table that should contain cards and whose values are the cards to render in that cell. The selector accounts for the current timeline view and whether any beats are collapsed.

Listing Card Map is an example of what this selector might produce.

NOTE: this selector does not produce full cards, but enough to render them on the timeline. Notably, it excludes custom attributes and the card description so that modifying a card doesn’t cause the entire timeline to re-render. i.e. as you modify one card’s description, this selector should continue to produce the same value (by pointer comparison).

- **visibleSortedBeatsForTimelineByBookSelector**: produces an array of the visible beats that should be rendered as columns in the order that they should be rendered. It should be used in the "default" and "tabbed" views, and takes into account the current view and whether any beats are collapsed.

Listing Visible Beats for Timeline is an example of what this selector might produce.

- **topTierBeatsInThreeTierArrangementSelector**: produces an array of the beats that should be rendered as the top bars in the stacked view of the timeline when there are three tiers. If there are only two tiers in the current structure, its result should be ignored when drawing the timeline table.

Listing Top Tier In Three Tier is an example of what this selector might produce.

- **secondTierBeatsInAtLeastTwoTierArrangementSelector**: produces an array of the beats that should be rendered as the second tier of bars in a structure with at least two levels when the timeline view is "stacked". If there are only two tiers, this will be the top row of bars.

Listing Second Tier in Two Tier is an example of what this selector might produce.

- **leavesPerBeat**: produces a Map of beat ids to the number of columns in width that it should span over in the "stacked" timeline view.

Listing Leaves Per Beat is an example of what this selector might produce. Note that in this example, beats 35 and 36 span 9 columns each, but all other beat titles don’t span more than one column.

3.1.1 Visible Beats for Timeline

```
[
  {
    "id": 35,
    "bookId": 7,
    "position": 0,
    "title": "auto",
    "time": 0,
    "templates": [],
    "autoOutlineSort": true,
    "fromTemplateId": null,
    "expanded": true
  },
  {
    "id": 34,
    "bookId": 7,
    "position": 0,
```

```

    "title": "auto",
    "time": 0,
    "templates": [],
    "autoOutlineSort": true,
    "fromTemplateId": null,
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 20,
    "position": 0,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 26,
    "position": 1,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 27,
    "position": 2,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 28,
    "position": 3,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 29,
    "position": 4,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,

```

```

    "fromTemplateId": null,
    "id": 30,
    "position": 5,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 31,
    "position": 6,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 32,
    "position": 7,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 33,
    "position": 8,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 36,
    "position": 1,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 38,
    "position": 0,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,

```

```

    "fromTemplateId": null,
    "id": 41,
    "position": 0,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 37,
    "position": 2,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 39,
    "position": 0,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 40,
    "position": 0,
    "time": 0,
    "title": "auto",
    "expanded": true
  }
]

```

3.1.2 Sparse Beat Map

```

{
  "0": 34,
  "1": 20,
  "2": 26,
  "3": 27,
  "4": 28,
  "5": 29,
  "6": 30,
  "7": 31,
  "8": 32,
  "9": 33
}

```

1. Card Map

```

{
  "13-33": [

```

```

{
  "id": 35,
  "beatId": 33,
  "lineId": 13,
  "tags": [],
  "color": null,
  "places": [],
  "characters": [],
  "bookId": null,
  "title": "Vah Ruta",
  "templates": [],
  "positionWithinLine": 0,
  "hasDescription": false
}
],
"13-32": [
{
  "id": 34,
  "beatId": 32,
  "lineId": 13,
  "tags": [],
  "color": null,
  "places": [],
  "characters": [],
  "bookId": null,
  "title": "Lynel",
  "templates": [],
  "positionWithinLine": 0,
  "hasDescription": false
}
],
"16-31": [
{
  "id": 33,
  "beatId": 31,
  "lineId": 16,
  "tags": [],
  "color": null,
  "places": [],
  "characters": [],
  "bookId": null,
  "title": "Zora's Domain",
  "templates": [],
  "positionWithinLine": 0,
  "hasDescription": false
}
],
"13-31": [
{
  "id": 32,
  "beatId": 31,
  "lineId": 13,
  "tags": [],
  "color": null,
  "places": [],
  "characters": [],
  "bookId": null,
  "title": "Prince Sidon",
  "templates": [],
  "positionWithinLine": 0,

```

```

        "hasDescription": false
    }
],
"13-30": [
    {
        "id": 31,
        "beatId": 30,
        "lineId": 13,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Making the climb",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"16-29": [
    {
        "id": 30,
        "beatId": 29,
        "lineId": 16,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Princess Ruto",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"13-29": [
    {
        "id": 29,
        "beatId": 29,
        "lineId": 13,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Zora's Domain",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"13-28": [
    {
        "id": 28,
        "beatId": 28,
        "lineId": 13,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],

```

```

        "bookId": null,
        "title": "Where to start",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"16-27": [
    {
        "id": 27,
        "beatId": 27,
        "lineId": 16,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "shadows",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"13-27": [
    {
        "id": 26,
        "beatId": 27,
        "lineId": 13,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Leaving the plateau",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"13-26": [
    {
        "id": 25,
        "beatId": 26,
        "lineId": 13,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Preparation",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"16-20": [
    {
        "id": 24,
        "beatId": 20,
        "lineId": 16,

```

```

        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "First memories",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"13-20": [
    {
        "id": 23,
        "beatId": 20,
        "lineId": 13,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Intro",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"14-24": [
    {
        "id": 22,
        "beatId": 24,
        "lineId": 14,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Level 3",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"14-23": [
    {
        "id": 21,
        "beatId": 23,
        "lineId": 14,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Level 2",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"14-22": [

```

```

    {
      "id": 20,
      "beatId": 22,
      "lineId": 14,
      "tags": [],
      "color": null,
      "places": [],
      "characters": [],
      "bookId": null,
      "title": "Level 1",
      "templates": [],
      "positionWithinLine": 0,
      "hasDescription": false
    }
  ],
  "14-21": [
    {
      "id": 19,
      "beatId": 21,
      "lineId": 14,
      "tags": [],
      "color": null,
      "places": [],
      "characters": [],
      "bookId": null,
      "title": "Intro - awake in a field and go in a cave",
      "templates": [],
      "positionWithinLine": 0,
      "hasDescription": true
    }
  ],
  "1-16": [
    {
      "id": 18,
      "beatId": 16,
      "lineId": 1,
      "tags": [],
      "color": null,
      "places": [],
      "characters": [],
      "bookId": null,
      "title": "Hyrule Castle",
      "templates": [],
      "positionWithinLine": 0,
      "hasDescription": false
    }
  ],
  "1-2": [
    {
      "id": 17,
      "beatId": 2,
      "lineId": 1,
      "tags": [],
      "color": null,
      "places": [],
      "characters": [],
      "bookId": null,
      "title": "Intro",
      "templates": [],
      "positionWithinLine": 0,

```

```

        "hasDescription": false
    }
],
"3-5": [
    {
        "id": 16,
        "beatId": 5,
        "lineId": 3,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "a scene",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": true
    }
],
"10-3": [
    {
        "id": 15,
        "beatId": 3,
        "lineId": 10,
        "tags": [],
        "color": null,
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "sub scene",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": true
    }
],
"6-11": [
    {
        "id": 6,
        "beatId": 11,
        "lineId": 6,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    },
    {
        "id": 13,
        "beatId": 11,
        "lineId": 6,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "another",
        "templates": [],
        "positionWithinLine": 1,
    }
]

```

```

    "hasDescription": false
  },
  {
    "id": 14,
    "beatId": 11,
    "lineId": 6,
    "tags": [],
    "places": [],
    "characters": [],
    "bookId": null,
    "title": "one more",
    "templates": [],
    "positionWithinLine": 3,
    "hasDescription": false
  }
],
"5-12": [
  {
    "id": 10,
    "beatId": 12,
    "lineId": 5,
    "tags": [],
    "places": [],
    "characters": [],
    "bookId": null,
    "title": "yeeee",
    "templates": [],
    "positionWithinLine": 0,
    "hasDescription": true
  },
  {
    "id": 11,
    "beatId": 12,
    "lineId": 5,
    "tags": [],
    "places": [],
    "characters": [],
    "bookId": null,
    "title": "andthis",
    "templates": [],
    "positionWithinLine": 1,
    "hasDescription": false
  },
  {
    "id": 12,
    "beatId": 12,
    "lineId": 5,
    "tags": [],
    "places": [],
    "characters": [],
    "bookId": null,
    "title": "scene3",
    "templates": [],
    "positionWithinLine": 3,
    "hasDescription": false
  }
],
"8-11": [
  {
    "id": 9,

```

```

        "beatId": 11,
        "lineId": 8,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "a thing",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"5-10": [
    {
        "id": 8,
        "beatId": 10,
        "lineId": 5,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "try again",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"6-12": [
    {
        "id": 7,
        "beatId": 12,
        "lineId": 6,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "anything",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"6-10": [
    {
        "id": 5,
        "beatId": 10,
        "lineId": 6,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "tuff",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"6-9": [
    {
        "id": 4,

```

```

        "beatId": 9,
        "lineId": 6,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"6-8": [
    {
        "id": 3,
        "beatId": 8,
        "lineId": 6,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"6-7": [
    {
        "id": 2,
        "beatId": 7,
        "lineId": 6,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
],
"5-6": [
    {
        "id": 1,
        "beatId": 6,
        "lineId": 5,
        "tags": [],
        "places": [],
        "characters": [],
        "bookId": null,
        "title": "Scene 1",
        "templates": [],
        "positionWithinLine": 0,
        "hasDescription": false
    }
]
}

```

```
[
```

```

{
  "id": 34,
  "bookId": 7,
  "position": 0,
  "title": "auto",
  "time": 0,
  "templates": [],
  "autoOutlineSort": true,
  "fromTemplateId": null,
  "expanded": true
},
{
  "autoOutlineSort": true,
  "bookId": 7,
  "fromTemplateId": null,
  "id": 20,
  "position": 0,
  "time": 0,
  "title": "auto",
  "expanded": true
},
{
  "autoOutlineSort": true,
  "bookId": 7,
  "fromTemplateId": null,
  "id": 26,
  "position": 1,
  "time": 0,
  "title": "auto",
  "expanded": true
},
{
  "autoOutlineSort": true,
  "bookId": 7,
  "fromTemplateId": null,
  "id": 27,
  "position": 2,
  "time": 0,
  "title": "auto",
  "expanded": true
},
{
  "autoOutlineSort": true,
  "bookId": 7,
  "fromTemplateId": null,
  "id": 28,
  "position": 3,
  "time": 0,
  "title": "auto",
  "expanded": true
},
{
  "autoOutlineSort": true,
  "bookId": 7,
  "fromTemplateId": null,
  "id": 29,
  "position": 4,
  "time": 0,
  "title": "auto",
  "expanded": true
}

```

```

    },
    {
      "autoOutlineSort": true,
      "bookId": 7,
      "fromTemplateId": null,
      "id": 30,
      "position": 5,
      "time": 0,
      "title": "auto",
      "expanded": true
    },
    {
      "autoOutlineSort": true,
      "bookId": 7,
      "fromTemplateId": null,
      "id": 31,
      "position": 6,
      "time": 0,
      "title": "auto",
      "expanded": true
    },
    {
      "autoOutlineSort": true,
      "bookId": 7,
      "fromTemplateId": null,
      "id": 32,
      "position": 7,
      "time": 0,
      "title": "auto",
      "expanded": true
    },
    {
      "autoOutlineSort": true,
      "bookId": 7,
      "fromTemplateId": null,
      "id": 33,
      "position": 8,
      "time": 0,
      "title": "auto",
      "expanded": true
    }
  ]

```

3.1.3 Top Tier In Three Tier

```

[
  {
    "id": 35,
    "bookId": 7,
    "position": 0,
    "title": "auto",
    "time": 0,
    "templates": [],
    "autoOutlineSort": true,
    "fromTemplateId": null,
    "expanded": true
  },
  {
    "autoOutlineSort": true,

```

```

    "bookId": 7,
    "fromTemplateId": null,
    "id": 36,
    "position": 1,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 37,
    "position": 2,
    "time": 0,
    "title": "auto",
    "expanded": true
  }
]

```

3.1.4 Second Tier in Two Tier

```

[
  {
    "id": 34,
    "bookId": 7,
    "position": 0,
    "title": "auto",
    "time": 0,
    "templates": [],
    "autoOutlineSort": true,
    "fromTemplateId": null,
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 38,
    "position": 0,
    "time": 0,
    "title": "auto",
    "expanded": true
  },
  {
    "autoOutlineSort": true,
    "bookId": 7,
    "fromTemplateId": null,
    "id": 39,
    "position": 0,
    "time": 0,
    "title": "auto",
    "expanded": true
  }
]

```

3.1.5 Leaves Per Beat

```

Map(17) {20 => 1, 26 => 1, 27 => 1, 28 => 1, 29 => 1,    }
[[Entries]]
0: {20 => 1}
1: {26 => 1}
2: {27 => 1}
3: {28 => 1}
4: {29 => 1}
5: {30 => 1}
6: {31 => 1}
7: {32 => 1}
8: {33 => 1}
9: {34 => 9}
10: {35 => 9}
11: {36 => 1}
12: {37 => 1}
13: {38 => 1}
14: {39 => 1}
15: {40 => 1}
16: {41 => 1}
size: 17
[[Prototype]]: Map

```

3.2 Relevant Actions

There are only a few ways to manipulate the beat tree. You may:

- add a beat (optionally specifying which beat should be the parent),
- insert a beat (specifying which beat should be it's sibling and inserting the beat at that level),
- expand a beat,
- collapse a beat, and
- append a top level beat.

3.2.1 Adding a Beat

`addBeat(bookId, parentId)`: lets you add a beat to the specified book. The `parentId` is optional. If you supply a `parentId` then the beat is created as a child of the supplied parent. If the `parentId` is omitted, then the beat is added as a top-level beat i.e. it's parent is `null`.

3.2.2 Inserting a Beat

`insertBeat(bookId, peerBeatId)`: lets you insert a beat to the right of the specified beat specified by `peerBeatId` in the book specified by `bookId`.

3.2.3 Expanding a Beat

`expandBeat(id, bookId)`: lets you specify a beat in a particular book to expand. This only impacts the "default" and "tabbed" views.

3.2.4 Collapsing a Beat

`collapseBeat(id, bookId)`: lets you specify a beat in a particular book to collapse. This will collapse the columns belonging to the specified beat's children into this beats column. This only impacts the "default" and "tabbed" views.

3.2.5 Appending A Top Level Beat

`appendTopLevelBeat(bookId)`: lets you append a beat to the top level of the timeline specified by `bookId`. i.e. this action cannot be used to create Chapters or Scenes in a three-tier structure.